

There is a problem with formula (9) which describes the formula for the D-matrix. In the following part of the formula are reconstructed from the R code.

Denom is the first part of formula (9)

$$\begin{aligned}
Denom^{-1} &= I \otimes I - \sum_i w_i(I \otimes H_{ii}) - \sum_i w_i(H_{ii} \otimes I) + \sum_i (K_{mi})^{\otimes 2} \sum_i w_i(HH_i)^{\otimes 2} \\
&= \sum_i w_i(I \otimes I) - \sum_i w_i(I \otimes H_{ii}) - \sum_i w_i(H_{ii} \otimes I) + \sum_i \sum_j (K_{mi})^{\otimes 2} w_j(HH_j)^{\otimes 2} \\
&= \sum_i w_i(I \otimes I) - w_i(I \otimes H_{ii}) - w_i(H_{ii} \otimes I) + \sum_i \sum_j w_j(K_{mi}HH_j)^{\otimes 2} \\
&= \sum_i w_i(I \otimes I - I \otimes H_{ii} - H_{ii} \otimes I) + \sum_i w_i(K_{mi}HH_i)^{\otimes 2} + \sum_i \sum_{j \neq i} w_j(K_{mi}HH_j)^{\otimes 2} \\
&= \sum_i w_i(I \otimes I - I \otimes H_{ii} - H_{ii} \otimes I + (H_{ii})^{\otimes 2}) + \sum_i \sum_{j \neq i} w_j(K_{mi}HH_j)^{\otimes 2} \\
&= \sum_i w_i(I - H_{ii})^{\otimes 2} + \sum_i \sum_{j \neq i} w_j H_{ij}^{\otimes 2}
\end{aligned}$$

with: $H_{ij} = K_{mi}HH_j$

$$vec(S_b) = \sum_i w_i b_i b'_i$$

$$\begin{aligned}
vec(c) &= vec(\sum_i R_i w_i - \sum_i H_{ii} R_i w_i - \sum_i R_i w_i H_{ii} + \sum_i K_i H H_i R_i w_i H H'_i K'_{mi}) \\
&= vec(\sum_i (R_i w_i - H_{ii} R_i w_i - R_i w_i H_{ii}) + \sum_i H_{ii} R_i w_i H'_{ii}) \\
&= \sum_i (vec(R_i w_i) - vec(H_{ii} R_i w_i) - vec(R_i w_i H_{ii})) + \sum_i vec(H_{ii} R_i w_i H'_{ii}) \\
&= \sum_i ((I \otimes I) vec(R_i w_i) - (I \otimes H_{ii}) vec(R_i w_i) - (H'_{ii} \otimes I) vec(R_i w_i) + (H'_{ii} \otimes H'_{ii}) vec(R_i w_i)) \\
&= \sum_i ((I \otimes I) - (I \otimes H_{ii}) - (H'_{ii} \otimes I) + (H'_{ii} \otimes H'_{ii})) w_i vec(R_i) \\
&= \sum_i (I - H_{ii})^{\otimes 2} w_i vec(R_i)
\end{aligned}$$

with: $R_i = \tilde{\Sigma} \otimes (Z'_i Z_i)^{-1}$ and H_{ii} a diagonal matrix

There are 2 problems:

- The code includes the weighting factor w_i . In the case where $w_i = \frac{1}{N_{clusters}}$ this would not matter, but it does matter for other weighting schemes.
- For $vec(c)$ the code misses $\sum_{k \neq i} H_{ik}^{\otimes 2}$